

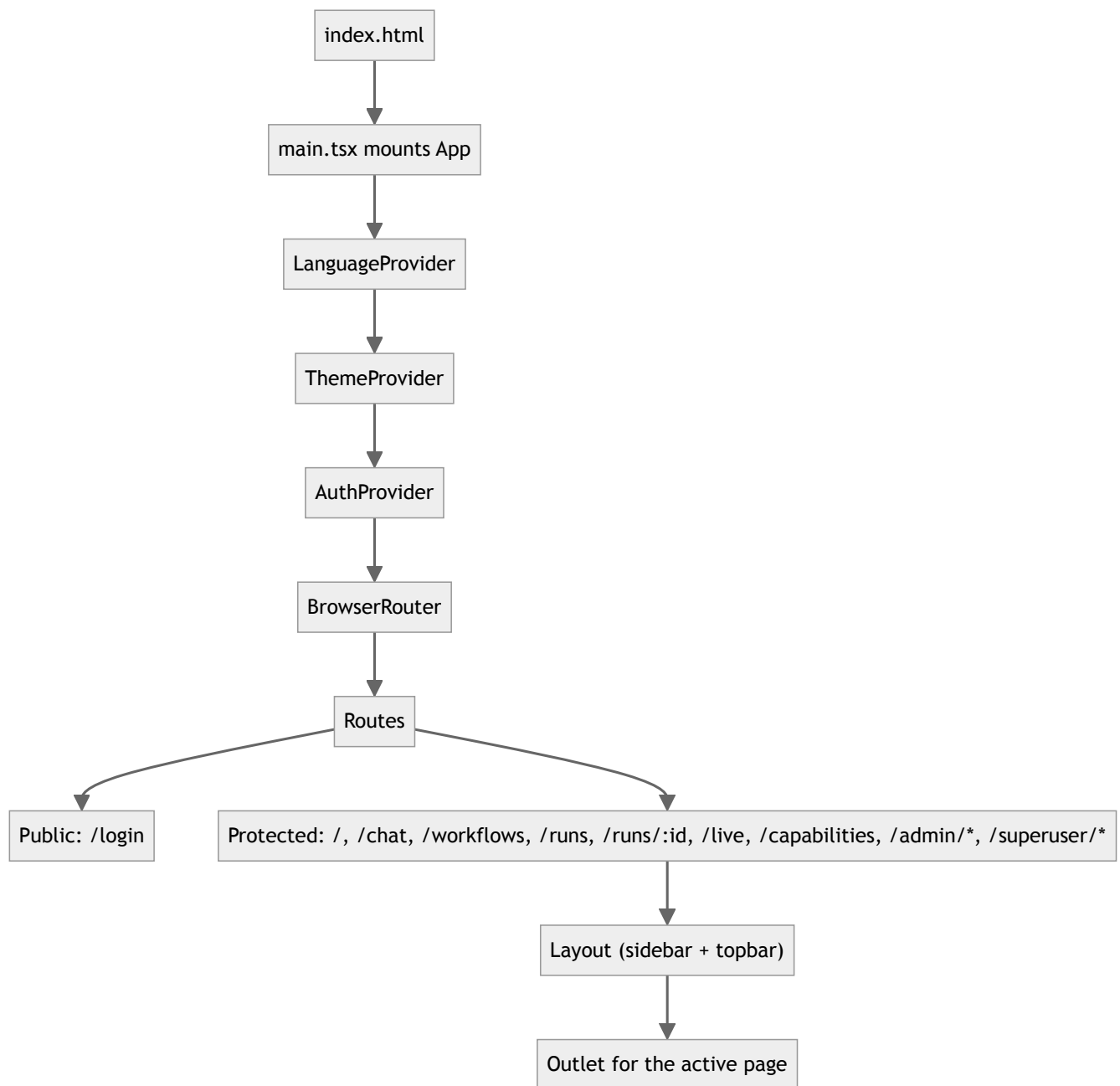
AAKAAR — Frontend Architecture (v1)

`aakar-web` is the single SPA that serves both superusers and tenant users. This document focuses on it. The mythic Sanskrit vocabulary (Pracharya, Mandala, Yajna, ...) is available as a UI language option but the design and engineering vocabulary throughout this doc is plain English.

1. Stack

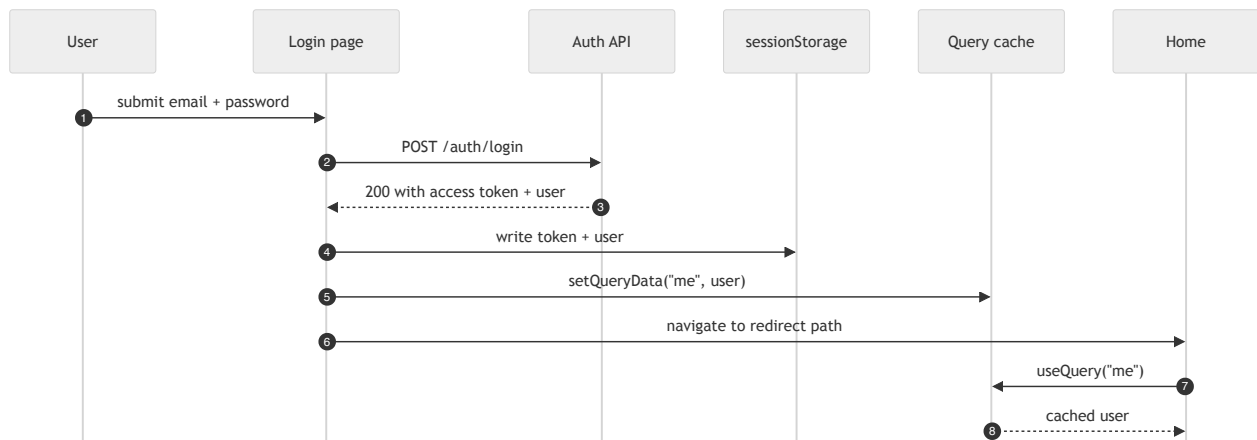
Concern	v1 choice
Build tool	Vite
Framework	React 18
Language	TypeScript
Routing	React Router 6
Server state	TanStack Query (React Query)
Forms	React Hook Form
Styling	Tailwind CSS
Graph	xyflow (React Flow) plus dagre
Charts	Recharts
Icons	lucide-react
Auth storage	sessionStorage (per tab)
i18n	tiny hand-rolled label map (<code>src/i18n/labels.ts</code>) — English default, five alternate languages
Themes	CSS-variable scoped per <code>[data-theme]</code> — 5 families × light/dark = 10 themes

2. App shell and routing



A `RequireAuth` wrapper guards the protected branch. If no token is present, it redirects to `/login` while preserving the intended location for post-login redirect.

3. Auth flow



Per-tab isolation is the reason for sessionStorage rather than localStorage: opening a second tab as a different role does not clobber the first tab's session.

4. API client

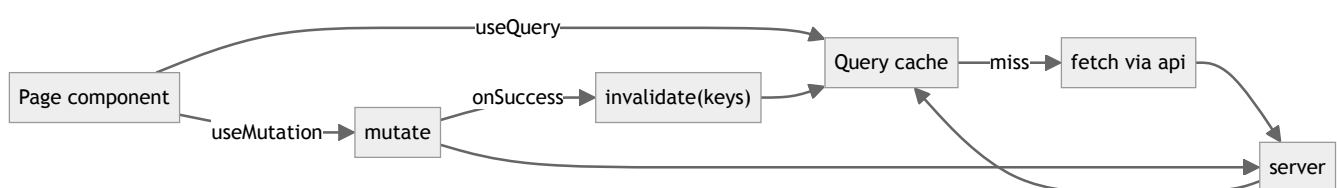
```

// src/api/index.ts (sketch)
const api = axios.create({ baseURL: "/api" });
api.interceptors.request.use(cfg => {
  const t = sessionStorage.getItem("aakar.token");
  if (t) cfg.headers.Authorization = `Bearer ${t}`;
  return cfg;
});
api.interceptors.response.use(
  r => r,
  err => {
    if (err.response?.status === 401) {
      sessionStorage.clear();
      location.replace("/login");
    }
    return Promise.reject(err);
  },
);

```

Every page imports typed wrappers (`api.runs.list` , `api.runs.get` , ...). Wrappers narrow request and response types to the shapes in `src/api/types.ts` . Wire field names stay English on both sides (`tenant_id` , `run.status` , etc.) so the backend contract is untouched.

5. TanStack Query patterns



Conventions:

- Query keys are arrays. First element is the resource: `["runs"]` , `["runs", id]` , `["chat", "sessions"]` .
- Mutations always invalidate the relevant list query on success rather than manually patching cache entries.
- SSE subscriptions for live run events are managed outside Query; they push into a Zustand-style local store keyed by run id.

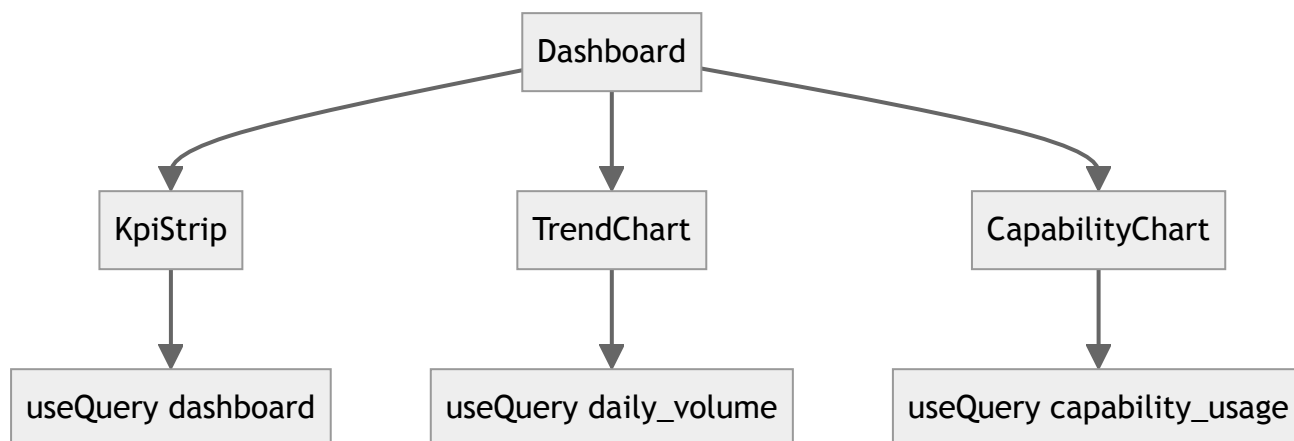
6. Pages

6.1 Dashboard

Lands at `/` . Three kinds of cards:

- **KPI strip** — runs last 24h, succeeded 7d, failed 7d, running right now.
- **Trend** — run volume over the last 30 days, stacked by status.
- **Capability usage** — top capabilities by invocation, failures highlighted.

Superusers additionally see a per-tenant stacked bar of the last 24 hours.

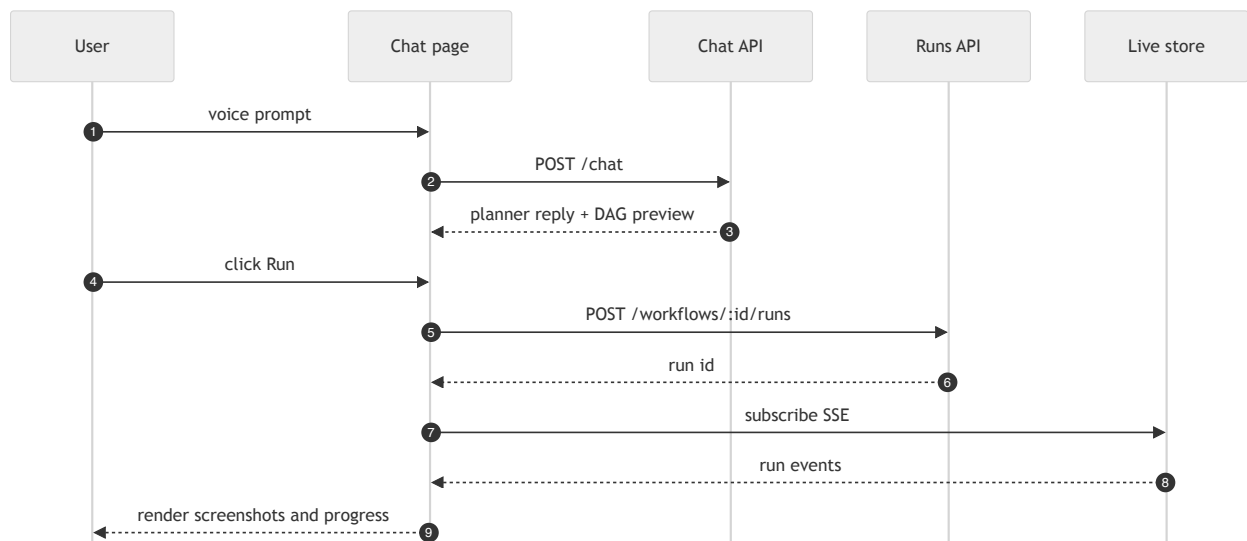


6.2 Chat

`/chat` is the user's primary surface. Layout:

- Left: list of chat sessions with a "New" button.
- Center: message thread with planner DAG previews inline.
- Right: live screen panel that activates when a run starts.

Long file paths and URLs in messages wrap on whitespace and `/` . Code-block detection prevents wrap inside fenced blocks.



6.3 Run detail (/runs/:id)

Shows the DAG view (xyflow + dagre auto-layout), an event timeline, and an artifacts list. Selecting a node scrolls the event timeline to its events and opens the latest screenshot.

6.4 Live processes

`/live` is a tenant-wide grid of in-flight runs. Each tile shows status, current node, and a thumbnail of the most recent screenshot. Tiles link to run detail.

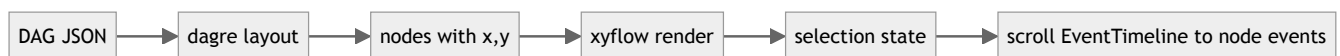
6.5 Vault

`/admin/grants` is a site-centric layout: the left rail lists sites the tenant has used; the right pane lists user handles registered for the selected site, with rotation hints. Only `tenant_admin` can edit.

7. Components inventory

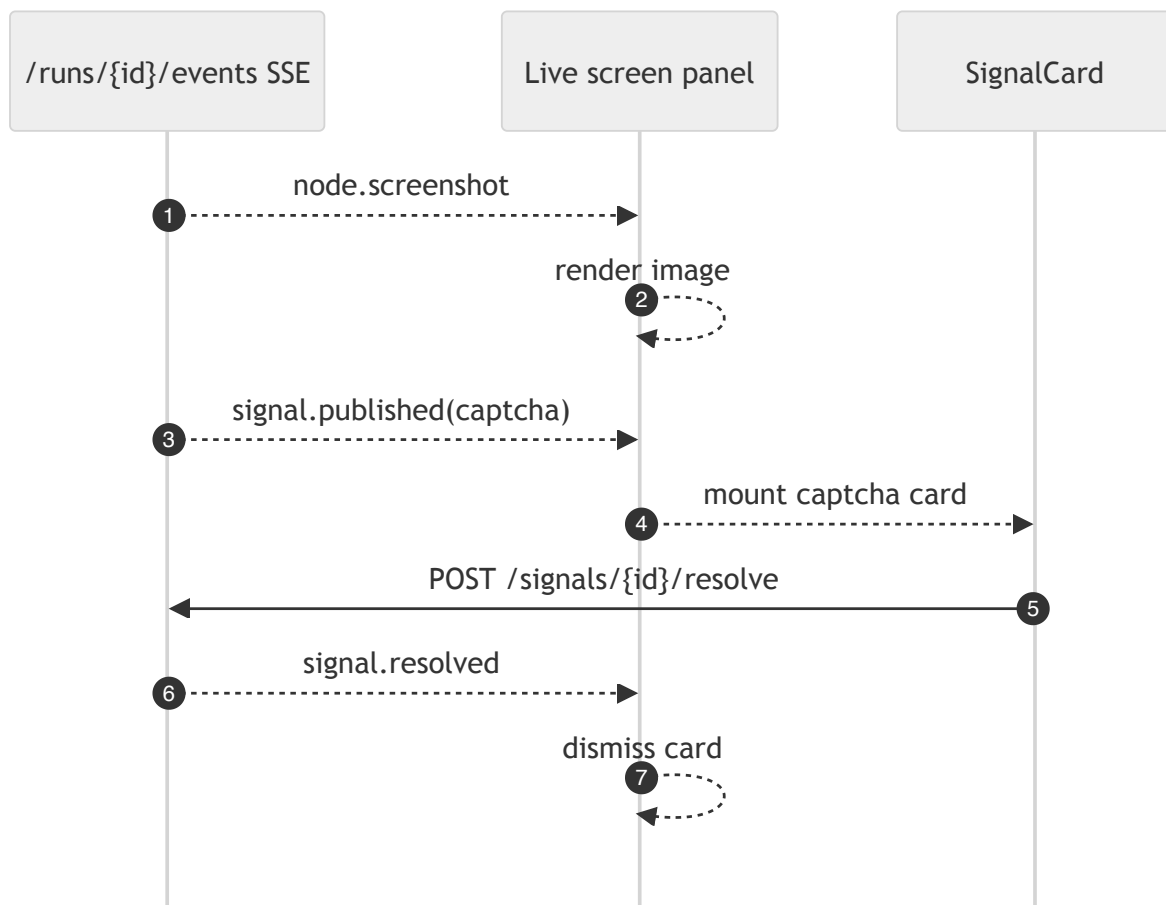
Component	Responsibility
Layout	shell with sidebar, topbar, and outlet
MorphLogo	branded logo
RequireAuth	route guard
MessageBubble	renders user or planner message with markdown and code
DagPreview	small read-only xyflow used in chat
DagView	full run-detail xyflow with selection
EventTimeline	scrollable run-event list with kind icons
LiveScreenPanel	screenshot stream plus signal handler
SignalCard	renders captcha, picker, or otp prompt and POSTs the resolution
IstTimestamp	formats ISO datetimes in IST
Toast	non-blocking notifications
LanguageSwitcher	sidebar popover for the i18n layer
ThemeSwitcher	sidebar popover for theme families
EasterEggs	tasteful client-side delights (Konami, type-word, triple-click)

8. DAG views



Node colors map to status: gray (queued), blue (running), green (succeeded), red (failed), amber (paused). Custom node renderers display the capability ref and elapsed time. `colorMode` and `<Background>` dot color follow the active theme via the `useTheme()` hook, so a light-mode session shows dark dots on white instead of cream-on-white invisibility.

9. Live screen panel



The panel debounces screenshot rendering (about 60 ms) so a fast-moving run does not flicker the canvas.

10. Dashboard charts

Recharts components are wrapped in a thin `Chart` boundary that handles loading and empty states. v1 uses three chart types: `BarChart` for capability usage, `LineChart` for daily run counts, and a small status-badge list for site health.

11. IST timestamps

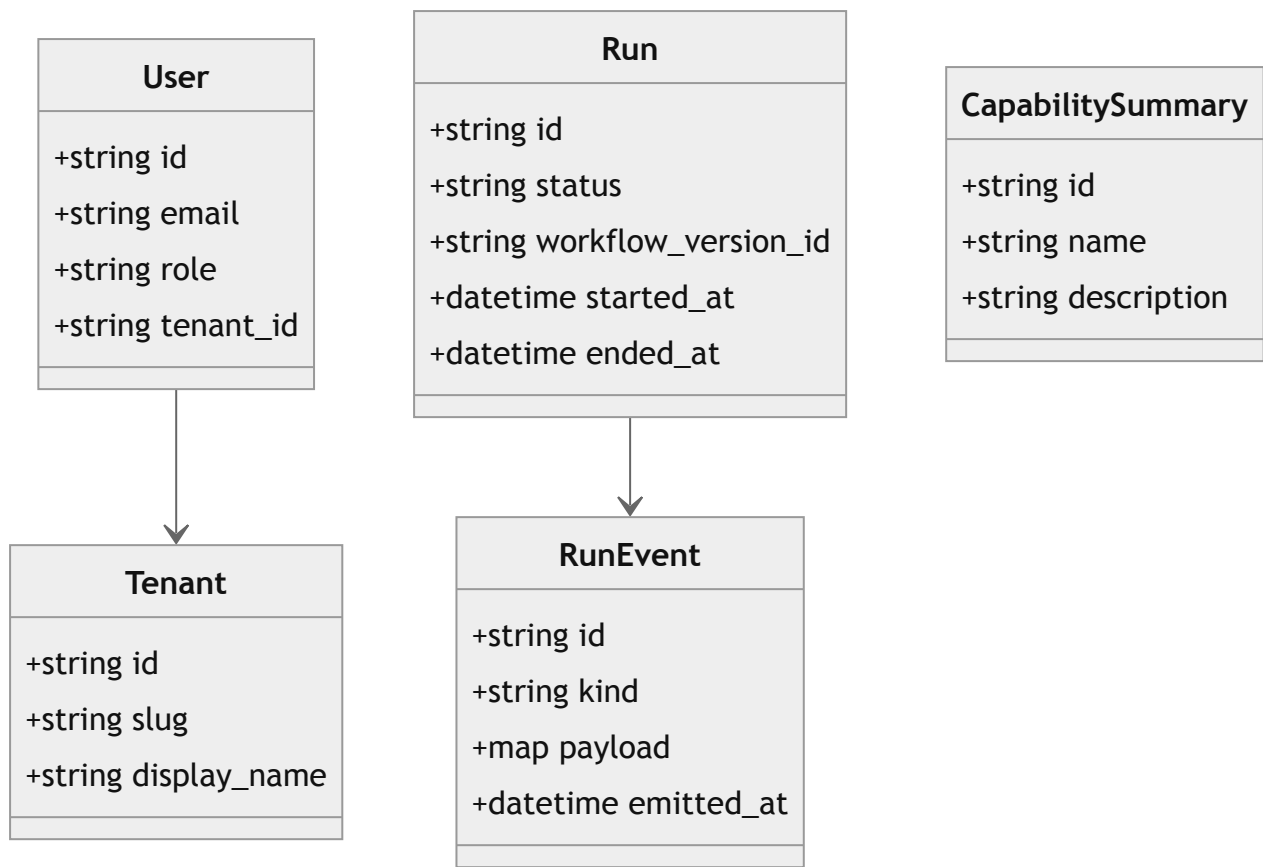
The backend stores UTC. The frontend formats in `Asia/Kolkata` for display because operators are in India and the audit log is read by ops in IST. The `IstTimestamp` component is the single source of truth for formatting; raw `Date.toLocaleString` is forbidden by lint.

12. i18n and themes

- **i18n** (`src/i18n/labels.ts`). Six languages: English (default), Hindi (Latin), Hindi (Devanagari), Bengali, Tamil, Kannada. Every operator-facing noun is a key in a single table. Pages call `useLabels()` and read `labels.tenant`, `labels.run`, etc. — never literals. Wire formats and code identifiers are unchanged across languages.

- **Themes** (`src/styles/themes.css` , `src/theme/ThemeProvider.tsx`). 5 families × light/dark = 10 themes, scoped via `[data-theme="<id>"]` . Components reference theme tokens (`bg-ink-950` , `text-accent-300`) and inherit the active palette. xyflow's pane background and edge strokes follow the active mode through `colorMode` + computed JS color props.

13. Type model



`src/api/types.ts` mirrors the backend's Pydantic shapes by hand. Drift between the two is caught in a small contract test; v1 has no codegen.

14. Build and deploy

Concern	v1
Local dev	<code>npm run dev</code> (Vite, port 5173)
Production build	<code>npm run build</code> (emits <code>dist/</code>)
Hosting	served as static files behind the same nginx that fronts the API
API base URL	<code>/api</code> (same origin), proxied by Vite in dev

The admin-app is deployed independently on its own subdomain (or port), with its own nginx and API service.

15. Conventions

- One component per file. File name matches the default export.
- Hooks live next to the component that uses them unless reused; reused hooks go to `src/hooks/`.
- Tailwind utility classes are preferred over CSS modules; CSS modules exist only for legacy admin-app pages.
- All forms go through React Hook Form. Manual `onChange` plumbing is forbidden by lint for any form field.
- Never put credentials or secrets in URLs, query params, or analytics events.
- All operator-facing nouns flow through `useLabels()`. Literal strings for nouns are forbidden by code review.

16. Reading guide

- For request and run lifecycles, read the backend doc.
- For the planner and capabilities, read the LLD.
- For where the UX is heading, read the roadmap.